Forecasting Swiss CPI inflation, 2026Q4–2029Q3

A bottom-up approach with an error correction model for oil products

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Abstract

We forecast Swiss year-on-year CPI inflation for 2026Q4–2029Q3 by splitting headline CPI into four blocks: oil products, rents, imported goods excluding oil and domestic goods and services excluding rents. Oil products follow an error correction model driven by Brent and the Swiss franc against the dollar; the other blocks follow univariate ARIMA models. Inflation is forecast at 0.9% in 2026Q4, falling to about 0.45% from mid-2027. A permanent 50% rise in Brent adds about 0.4 percentage points for one year. Over 55 recursive forecast origins since 2012, the forecasts are unbiased and beat a headline ARIMA and a random walk at horizons of two to three years, but they are not efficient: beyond one year they miss turning points.

1 Introduction

The head of the SNB's forecasting team is mainly concerned that an oil-price shock feeds into Swiss inflation. We therefore model headline CPI bottom-up and put the exogenous drivers, Brent and the CHF/USD exchange rate, in the oil block. Section 2 describes the data, Section 3 the models and their optimal forecasts, Section 4 the forecast, Section 5 the out-of-sample evaluation and Section 6 the limitations and next steps. *[To be completed: motivation, SNB price-stability range, related literature (e.g. Huwiler and Kaufmann, 2013).]*

2 Data and blocks

Monthly CPI indices and basket weights come from the Federal Statistical Office (BFS), Brent from FRED and CHF/USD from the SNB. Monthly series are converted to quarterly averages. The estimation sample is 2000Q1–2026Q3 (107 quarters). For each block we use $y_t = 100 \ln(\text{index}_t)$, so Δy_t is quarter-on-quarter inflation in percent and $\pi_t = y_t - y_{t-4}$ is year-on-year inflation.

The BFS publishes imported and domestic aggregates but not the blocks excluding oil and rents. We build them within each annual chain link (index relative to the previous December), assuming all oil products are imported and all rents domestic, and chain the links. Weights before 2016 are recovered from the CPI identity by least squares; against the published 2016–2026 weights the error is at most 0.04 pp.

Re-aggregation. Rebuilding headline CPI from the four blocks, linked at Q4 with each year's weights, reproduces published y-y inflation within 0.14 pp (maximum, 2009) and 0.02 pp on average since 2000. *[To be completed: descriptive statistics and figure of the four blocks.]*

Table 1: Blocks, 2026 basket weights and models

Block	Weight	Model	Drivers
Domestic excl. rents	59.0%	ARIMA$(0,1,0)(1,0,0)_4$ with drift	–
Imported excl. oil	20.1%	ARIMA$(2,1,1)(1,0,0)_4$ with drift	–
Rents	18.9%	ARIMA$(1,1,1)(1,0,0)_4$ with drift	–
Oil products	2.1%	Error correction model	Brent (USD), CHF/USD, trend

3 Models and optimal forecasts

Under quadratic loss the optimal h-step forecast is the conditional expectation given the information at the origin T = 2026Q3: future shocks are set to their expected value of zero and observed values are kept.

3.1 Oil products: error correction model

The long-run relation, estimated by OLS (Engle–Granger first step), is

$$oil_t = 247.58 + \underset{(0.014)}{0.428}\, brent_t + \underset{(0.054)}{0.614}\, usd_t + \underset{(0.027)}{0.415}\, t + ect_t, \tag{1}$$

where *brent* is 100 ln Brent in USD and *usd* is 100 ln CHF per USD. A 10% rise in Brent raises oil-product prices by about 4.3% in the long run; the elasticity is below one because taxes and margins are fixed per litre. The short-run dynamics (one lag, chosen by BIC) are

$$\begin{aligned}\Delta oil_t = \underset{(0.27)}{0.398} - \underset{(0.091)}{0.394}\; ect_{t-1} + \underset{(0.017)}{0.309}\, \Delta brent_t + \underset{(0.078)}{0.539}\, \Delta usd_t \\ + \underset{(0.077)}{0.017}\, \Delta oil_{t-1} + \underset{(0.037)}{0.095}\, \Delta brent_{t-1} - \underset{(0.090)}{0.012}\; \Delta usd_{t-1} + \varepsilon_t.\end{aligned} \tag{2}$$

About 39% of a gap to equilibrium closes each quarter. All three variables are I(1) (ADF with trend). Engle–Granger finds cointegration ($-5.30 < -4.10$ at 5%); the Johansen trace test does not reject $r = 0$ ($37.98 < 42.44$), but its first cointegrating vector (0.44, 0.67, 0.44) is close to the Engle–Granger estimates. Equal Brent and USD elasticities are rejected ($F = 16.7$, $p = 0.0001$).

Optimal forecast. The three equations are iterated forward for $h = 1, \ldots, 12$:

$$\begin{aligned}\widehat{\Delta oil}_{T+h} = c + \alpha\, \widehat{ect}_{T+h-1} + b_0 \widehat{\Delta brent}_{T+h} + g_0 \widehat{\Delta usd}_{T+h} \\ + a_1 \widehat{\Delta oil}_{T+h-1} + b_1 \widehat{\Delta brent}_{T+h-1} + g_1 \widehat{\Delta usd}_{T+h-1},\end{aligned} \tag{3}$$

$$\widehat{oil}_{T+h} = \widehat{oil}_{T+h-1} + \widehat{\Delta oil}_{T+h}, \tag{4}$$

$$\widehat{ect}_{T+h} = \widehat{oil}_{T+h} - \beta_0 - \beta_1 \widehat{brent}_{T+h} - \beta_2 \widehat{usd}_{T+h} - \delta\,(T+h). \tag{5}$$

Because Brent and CHF/USD enter contemporaneously, the forecast is conditional on their paths. We use random walks, so their future changes are zero. Oil inflation then converges to the trend $\delta = 0.415$ per quarter (about 1.7% a year), with *ect* settling at $(\delta(1-a_1)-c)/\alpha \approx -0.03$.

3.2 Rents, imported and domestic goods: ARIMA with drift

Each block follows

$$\phi(L)\, \Phi(L^4)\, (\Delta y_t - \mu) = \theta(L)\, \varepsilon_t, \tag{6}$$

with drift μ, the mean q-q inflation rate.

Table 2: ARIMA estimates, 2000Q1–2026Q3 (standard errors in brackets)

	Rents $(1,1,1)(1,0,0)_4$	Imported excl. oil $(2,1,1)(1,0,0)_4$	Domestic excl. rents $(0,1,0)(1,0,0)_4$
ϕ_1	0.329 (0.138)	−0.701 (0.111)	
ϕ_2		0.273 (0.105)	
θ_1	0.529 (0.140)	0.736 (0.079)	
Φ_1	0.272 (0.101)	0.514 (0.093)	0.725 (0.066)
μ	0.357 (0.058)	−0.140 (0.167)	0.149 (0.083)
Ljung–Box p (8 lags)	0.74	0.64	0.11
BIC	−21.0	261.3	30.0

Optimal forecast. Write $a(L) = \phi(L)\Phi(L^4) = 1 - \sum_{i=1}^{P} a_i L^i$ and $x_t = \Delta y_t - \mu$. Then

$$\hat{x}_{T+h} = \sum_{i=1}^{P} a_i \, \hat{x}_{T+h-i} + \theta_1 \hat{\varepsilon}_T \mathbb{1}[h = 1], \tag{7}$$

$$\hat{y}_{T+h} = y_T + \sum_{j=1}^{h} (\mu + \hat{x}_{T+j}), \qquad \hat{\pi}_{T+h} = \hat{y}_{T+h} - \hat{y}_{T+h-4}, \tag{8}$$

where $\hat{x}_{T+h-i} = x_{T+h-i}$ for dates up to T. As $\hat{x}$ dies out, $\hat{\pi}$ converges to 4μ. With $\Delta y_t - \mu = \psi(L)\varepsilon_t$ and cumulated weights $\Psi_k = \sum_{i=0}^{k} \psi_i$ ($\Psi_k = 0$ for $k < 0$), the y-y forecast error variance is

$$\mathrm{Var}(\pi_{T+h} - \hat{\pi}_{T+h}) = \sigma^2 \sum_{k=0}^{h-1} (\Psi_k - \Psi_{k-4})^2 . \tag{9}$$

Our coded recursion reproduces R's `predict()` to 10^{-13}. In the imported model the AR inverse root −0.98 nearly cancels the MA root −0.74, a sign of over-parameterisation.

3.3 Aggregation and density

Block forecasts are converted to index levels and aggregated with the 2026 weights, linked at Q4: $P_t^{CPI} = P_{Q4,t-1}^{CPI} \sum_i w_i P_t^i / P_{Q4,t-1}^i$. The density forecast uses 5,000 simulated paths; in each future quarter we draw one past quarter and take, from that quarter, the residuals of all four models and the changes in Brent and CHF/USD, which keeps their correlation. Parameter uncertainty is not included.

4 Forecast

Headline inflation is forecast at 0.93% in 2026Q4, a result of the 2026 oil spike (Brent rose from \$64 in 2025Q4 to \$103 in 2026Q2). As the spike drops out of the y-y rate and oil products return towards equilibrium, inflation falls to 0.22% in 2027Q3 and then settles around 0.45% (Figure 1). In the long run, rents contribute +0.27 pp, domestic goods +0.28 pp, imported goods −0.13 pp and oil products +0.03 pp. A permanent 50% rise in Brent from 2027Q1 adds about 0.4 pp for four quarters; it raises the price level permanently but y-y inflation only for one year.

5 Out-of-sample evaluation

We produce recursive (expanding-window) forecasts from 55 origins, 2012Q4–2026Q2, for $h = 1, \ldots, 12$, re-estimating all models at each origin (Figure 2). Tests follow the lecture (Section 3.1), with Newey–West standard errors using $h - 1$ lags.

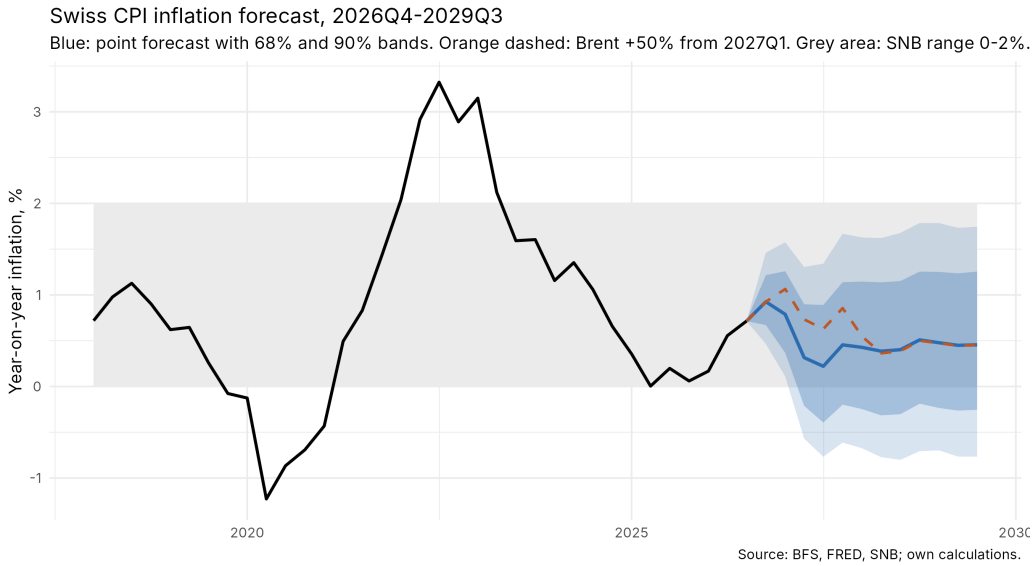


Figure 1: Headline CPI inflation forecast with 68% and 90% bands, and the Brent +50% scenario

Table 3: Absolute standards (p-values unless stated)

	$h = 1$	$h = 4$	$h = 8$	$h = 12$
Mean error (pp)	0.02	0.04	0.08	0.25
RMSE (pp)	0.36	1.06	1.29	1.20
Unbiasedness, $H_0 : \tau = 0$	0.71	0.90	0.84	0.50
Mincer–Zarnowitz, $H_0 : \alpha = 0, \beta = 1$	0.65	0.36	0.006	0.000
$\hat{\beta}$	1.06	0.64	−0.54	−0.53
Strong efficiency, $H_0 : \beta = 0$ on Ω_t	0.013	0.038	0.005	0.000
Weak efficiency, errors at most MA($h-1$)	0.015	0.001	0.001	0.24

The forecasts are unbiased at all horizons (Table 3). The bottom-up model beats the headline ARIMA by 20–25% in MSFE at $h = 8$–12 and the random walk at all horizons (Table 4), so the disaggregation adds value at medium horizons. Beyond one year, however, Mincer–Zarnowitz and both efficiency tests reject and the model does no better than the sample mean: it extrapolates the current situation and misses turning points, such as the 2022 surge and the 2024–25 decline. MAFE ratios give the same ranking. *[To be completed: density evaluation (coverage, PIT), error variance by horizon.]*

6 Limitations and next steps

- **Seasonality.** The ARIMA models contain seasonal AR terms. Next: seasonality plots by quarter, X-13 adjustment where seasonal, plain ARIMA.
- **Persistence beyond one year.** Next: test deterministic against stochastic trends for domestic goods (ADF with trend).
- **Imported block.** The negative drift (−0.14, not significant) lowers headline inflation by 0.13 pp over the whole horizon, and the AR and MA roots nearly cancel. Next: drop the drift or add a 2009–2016 dummy; estimate ARIMA(1,1,0) or (0,1,1).
- **Density bands.** The 90% half-width at $h = 8$ is 1.24 pp, against about 2.1 pp implied by the out-of-sample RMSE. Next: check coverage and calibrate on out-of-sample errors.

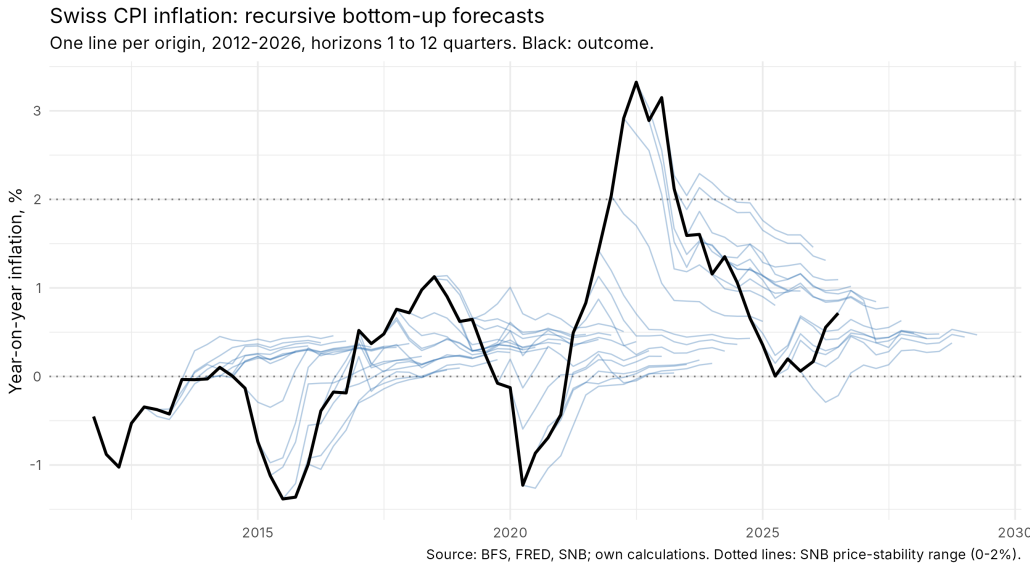


Figure 2: Recursive bottom-up forecasts against the outcome

Table 4: Relative standards: MSFE ratio, model over benchmark (Diebold–Mariano p, HLN correction)

Benchmark	$h = 1$	$h = 4$	$h = 8$	$h = 12$
ARIMA on headline CPI	0.99 (0.97)	1.04 (0.80)	0.80 (0.04)	0.75 (0.04)
Random walk on y-y inflation	0.78 (0.29)	0.84 (0.39)	0.56 (0.02)	0.46 (0.07)
Sample mean	0.11 (0.00)	0.85 (0.31)	1.17 (0.35)	1.11 (0.47)

- **Outliers.** COVID and 2022: additive outliers where a residual exceeds three times the interquartile range.

Questions for the meeting of 4 November

1. Is X-13 seasonal adjustment followed by a plain ARIMA acceptable for the non-oil blocks?
2. Engle–Granger finds cointegration and Johansen does not: is Engle–Granger enough to keep the ECM?
3. Can we extrapolate the ECM's linear trend for 12 quarters, or should CO_2-levy dummies replace it?
4. Brent and CHF/USD: random walk plus a shock scenario, or the futures curve?
5. Imported block: drop the drift, add a 2009–2016 dummy, or start the sample later?
6. Are the least-squares weights before 2016 (maximum error 0.04 pp) acceptable?

A Code and reproducibility

All results come from one R project. Running `source("main.R")` reproduces every table and figure in about 30 seconds. Parameters (sample start, horizon, model orders, first evaluation origin, number of simulations, scenario) are set once at the top of `main.R`. Raw data are downloaded through APIs (BFS, FRED, SNB), never edited by hand, and all paths are relative (`here()`).

```
Forecasting/
|-- main.R                        parameters + runs everything in order
|-- R/functions.R                 download helpers
|-- R/models.R                    model, forecast, aggregation and test functions
|-- Scripts/01_load_data.R        raw data from the APIs into RAW_DATA/
|-- Scripts/02_dta_cleaning.qmd   cleaning, block indices, weights, quarterly data
|-- Scripts/04_model_estimation.R  estimates, diagnostics, cointegration, checks
|-- Scripts/05_forecasting.R      point and density forecast, scenario, fan chart
|-- Scripts/06_evaluation.R       recursive forecasts, evaluation tables, charts
|-- DTA_CLEAN/  output/  figures/  report/
```

Estimation uses transparent functions only (`stats::arima` by maximum likelihood, `lm`, `urca`); no automatic model selection. The forecasts are coded by hand from the derivations in Section 3, so each equation has a matching function (Table 5).

Table 5: Main functions in `R/models.R`

Function	What it does	Implements
`fit_arima()`	ARIMA$(p,1,q)(P,0,Q)_4$ with drift, by ML	eq. (6)
`arima_polys()`	Multiplies out $a(L) = \phi(L)\Phi(L^4)$ and the MA side	eq. (7)
`arima_paths()`	Iterates the ARIMA forward, back to log levels; zero shocks give the point forecast, drawn shocks give simulated paths	eq. (7)–(8)
`fit_ecm()`	Engle–Granger two steps: long-run OLS, then short-run OLS with ect_{t-1}	eq. (1)–(2)
`ecm_paths()`	Iterates the three ECM equations forward, given driver paths	eq. (3)–(5)
`aggregate_headline()`	Rebuilds headline y-y inflation from block levels, linked at Q4	Section 3.3
`forecast_point()`	Fits all four blocks at one origin and returns the headline forecast	–
`wald_hac()`	Wald test with Newey–West covariance ($h-1$ lags): MZ, unbiasedness, strong efficiency	Table 3
`weak_efficiency_p()`	Ljung–Box on the residuals of an MA$(h-1)$ fitted to the errors	Table 3
`dm_test()`	Diebold–Mariano with the Harvey–Leybourne–Newbold correction	Table 4

Check of the derivation: with zero shocks, `arima_paths()` returns the same forecasts as R's Kalman-filter `predict()`, with a maximum difference of 10^{-13} for all three blocks. The core of the recursion, steps 2 and 3 of eq. (7)–(8):

```r
for (j in 1:h) {
  t <- n_x + j
  ar <- colSums(pl$a * X[t - seq_len(P), , drop = FALSE]) # sum of a_i * x_{T+h-i}
  ma <- colSums(pl$m * E[t - seq_len(Q), , drop = FALSE]) # MA terms: eps_T only at h = 1
  X[t, ] <- ar + E[t, ] + ma # future shocks E = 0 (point)
}
dy <- X[n_x + 1:h, , drop = FALSE] + pl$mu # back to q-q inflation
y[length(y)] + matrix(apply(dy, 2, cumsum), h, N) # cumulate to log levels
```